



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 14-300-1

Date: 01-03-2022

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**TECHNICAL SPECIFICATIONS
FOR
LOW VOLTAGE HEAT SHRINKABLE CABLE JOINTS
0.6/1(1.2) kV**

Issue: Mar-2022/ Rev- 1

توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.
يلزم إرفاق المواصفة الخاصة بالسرافيل (EDMS 15-301-1 (Compression Type Connectors)



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1. SCOPE

This specification covers the minimum technical requirements for design, engineering, manufacture, testing, inspection, installation and performance of all kinds of straight through joints for low voltage power cables of 1 kV level system, XLPE insulation, 0.6/1 (1.2) kV with the following different constructions and sizes:

- Single/Three cores with neutral core.
- Armoured and non armoured.
- Copper and aluminum conductors.

2. APPLICABLE STANDARDS

The low voltage heat shrinkable joints should be complying with the latest edition of IEC&DIN publications unless otherwise specified in this specification. The equipment/material covered in this specification shall comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from those of the standards listed below at table (1), the provisions of this specification shall apply. Unless otherwise specified Joints should comply with all relevant standards described in table (1).

TABLE 1: Applicable Standards Description

<u>S. N.</u>	<u>Standard No.</u>	<u>Description</u>
1	IEC 60502-1	Power cables with extruded insulation and their acc. For rated voltages from 1 KV ($U_m=1.2$ KV and 3 KV ($U_m = 3.6$ KV)
2	DIN 46 267-1	non-tension proof compression joints for copper conductors (CU connectors)
3	DIN 46 267-2	non-tension proof compression joints for Aluminum conductors (AL connectors)
4	EDMS 03-300-2	Underground XLPE cables 0.6/1(1.2) kV Aluminum/Copper, Armoured
5	EDMS 15-301-1	Compression Type Connectors for Cables up to 24 kV
6	IEC 60228	Conductors of insulated cables
7	ASTMD 792, 573, 2240 ISO 527, 868	Heat Shrinkable tubes



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3. ENVIRONMENTAL CONDITIONS

The performance of the low voltage heat-shrinkable joint should be guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Minimum ambient temperature	: -5°C.
Maximum ambient temperature	: 45°C (50 °C as option).
Maximum relative humidity	: 95%.
Maximum altitude	: 1000 m.
Soil conditions	: Exposed to sea water / sewage water, and Soil conditions vary from dry sand to wet silt or clay.

4. SYSTEM DETAILS

Frequency	50 Hz
Number of phases	Single/ multi core
Nominal voltage	0.4 KV
Highest voltage	0.6 KV
Maximum S.C. temperature	250°C

5. SERVICE CONDITIONS

- The sunlight is very intense and the temperature of exposed 70°C.
- The maximum solar radiation intensity on a horizontal surface is 1000 w/m².
- Severe surface contamination from salt, dust, heavy industrial pollution, sandstorms, fine dust often carried in suspension in the atmosphere even on calm days, can be expected.
- Ground temperature can reach 35°C.
- Joints should be suitable for direct burial on the following conditions: -
 - a) Where the water-table / sea water / sewage water.
 - b) With laying depth of the cable up to 1.0 m.



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6. JOINT MATERIALS, RATING AND PERFORMANCE

6.1 The minimum guaranteed performance should be under Egypt service conditions mentioned in item (5).

6.2 The low voltage heat shrinkable cable joints should be suitable for use with the following type of cables:

- 0.6/1 (1.2) kV, three core, underground power cable, Aluminum/ Copper stranded-compacted sectorial conductor, XLPE insulation, filler, separation sheath ST2 (bedding), double Steel tape armour, black PVC over sheath, and according to the requirements of IEC 60502-1. This specification covers the different joints of underground cables buried directly with cross section areas as follow table (2):

TABLE 2: Cross Section Area of Under Ground Cables

4x25 mm ²
3x35 + 16 mm ²
3x50 + 25 mm ²
3x70 + 35 mm ²
3x95 + 50 mm ²
3x120 + 70 mm ²
3x150 + 70 mm ²
3x185 + 95 mm ²
3x240 + 120 mm ²
3x300 + 150 mm ²
3x400 + 240 mm ²
3x500 + 240 mm ²

6.3 The materials of the joints and the installation instructions should be compatible with the XLPE cable materials 0.6/1 (1.2) kV and specification as:

- a- Conductors: Aluminum / copper stranded sectorial conductors.
- b- Insulation: XLPE rated 90°C for continuous operation.
- c- Inner covering: a layer of extruded PVC.
- d- Armour: two steel tapes applied helically over the inner covering.



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- e- Sheath: PVC rated 70°C colour black.
- f- Standards: Conform with IEC 60502-1.
- g- System service voltage: 380 / 220 V, 50 HZ.

6.4 Connectors material, outer diameter, die size as per DIN 46267 part-1 for copper connector and as per part-2 for aluminum connectors.

6.5 Connectors length as per joint manufacturers design and joint type on their own responsibility.

6.6 Connectors diameters, die size, and tolerances should be as the following tables:

TABLE 3: Aluminum Connector diameters and crimping size

C.S.A (mm ²)	Connector inner diameter mm	Connector outer diameter mm	Crimping die size
16	6.5	12	12
25	6.8	12	12
35	8.0	14	14
50	9.8	16	16
70	11.2	18	18
95	13.2	22	22
120	14.7	22	22
150	16.3	25	25
185	18.5	28	28
240	21.0	32	32
300	23.3	34	34
400	26.0	38	38
500	29.0	44	44

Note:

- * The above dimensions are subjecting for a manufacturing tolerance upto +/- 0.5mm.
- * The recommended die is the hexagonal type, with hydraulic crimping tool.



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- * The crimping area of the connector should not be less than 50% of the connector total length.
- * The sector shape conductors should be pre-rounding before crimping.

TABLE 4: Copper Connectors diameters and crimping size

Copper connector for low voltage cables			
C.S.A (mm ²)	Connector inner diameter (mm)	Connector outer diameter (mm)	Crimping die size
16	5.5	8.5	8
25	7	10	10
35	8.2	12.5	12
50	10	14.5	14
70	11.5	16.5	16
95	13.5	19	18
120	15.5	21	20
150	17	23.5	22
185	19	25.5	25
240	21.5	29	28
300	24.5	32	32
400	27.5	38.5	38
500	31	42	42

Note:

- * The above dimensions are subjecting for a manufacturing tolerance up to +/- 0.5mm.
- * The recommended die is the hexagonal type, with hydraulic crimping tool.
- * The crimping area of the connector should not be less than 50% of the connector total length.
- * The sector shape conductors should be pre-rounding before crimping.



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6.7 Connectors data sheets for joints complying with this specification:

TABLE 5: Aluminum Connector Data Sheet

Description	Aluminium Connectors
Application	For Stranded Aluminum Compacted Conductors Complying to IEC 60228
Material	One piece seamless tube, EN Al 1050 (AL99.5) According to DIN EN755-7
Finishing	Chemical Treatment
Identification	Manufacturer name, conductor size, die size, number of crimping, and crimping positions
Dimention and codeing	As per drawing and table of dimentions
Mechanical proparties	Pure and soft AL
Chemical composition	Min. AL ratio 99.5% (mass friction)
Electrical conductivity	Min. conductivity 56% according to IACS. IACS: International annelid copper standard

TABLE 6: Copper Connector Data Sheet

Description	Copper Connectors
Application	For Stranded Copper Compacted Conductors Complying to IEC 60228
Material	One piece seamless tube, Electrolytic tough pitchCopper (Cu-ETP) according to DIN EN 13600 (supersedes DIN 40500)
Finishing	Tinned coating not less than 3 micro m
Identification	Manufacturer name, conductor size, die size, number of crimping, and crimping positions
Dimentions and codeing	As per drawing and table of dimentions
Mechanical proparties	Pure Copper
Chemical composition	Min. Copper ratio 99.9% (mass friction)
Electrical conductivity	Min. conductivity = 96.6% according to IACS. IACS: International anneled copper standered



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6.8 Each joint shall be complete with the necessary and suitable ferrules for AL cables each ferrule should be complete with the inhibitor and two plastic caps as covering.

6.9 Each splice kit shall be suitable to make jointing between cables with cables over sheath made of PVC.

6.10 The Joints kits should be supplied completely with all relevant components including the heat shrinkable insulation tubes for cable cores, DIN ferrules (connectors), tinned copper Braid and clamps for armouring (as applicable), outer jacket tubes, and/or any special accessories required.

6.11 Every kit shall consist of the following: -

- a- Outer tubing (shrinkable with adhesive).
- b- Four insulating tubing for connectors (shrinkable with adhesive).
- c- Tubular Tinned Cu braid.
- d- Galvanized steel clamps: 2 pcs for armoring cables only (to bind the Cu Braid with cable armoring), plastic and steel ties are not applicable.
- e- Four connectors (in case of AL connectors an inhibitor and two plastic caps should be supplied).
- f- Sealant strip: Black Mastic tapes (pastes) of types required for sealing and filling, as applicable at least Over steel clamps (if exist), under the ends of jacket tube over the cables.
- g- One insulating welding PVC tape.
- h- Installation instructions including the Kit contents list.

7. MARKING AND IDENTIFICATION

All components of the joint should be distinctly and clearly marked with the manufacturer's name and reference numbers. The marking should be done before packing the component.

Components or materials, if subjected to a shelf-life limitation, should have the final date of use prominently and permanently shown on all packages.

The electrical relevant components of cable accessories must be marked with the following data:



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1. Manufacturer name.
2. Production indicator for traceability.
3. Cross-section, or diameter, or application range should be marked.

8. PACKING

8.1. Packing should be robust and able to withstand very rough handling during transportation. It should be protected against ingress of moisture.

8.2. The details of method of packing should be submitted with the offer.

8.3. Packaging as well for each cable accessory should be printed and possible to read all information directly with the following information: -

- a) Manufacturer's name.
- b) Manufacturer's Type No. / catalog number.
- c) Purchase order number/tender.
- d) Cable size/diameter.
- e) Nominal Voltage (0.6/1 kV) / Max. Voltage (1.2 kV)
- f) Year of manufacture
- g) Customer Name EDC
- h) Manufacturer's batch No. or any code for tracing the quality assurance records maintained by the manufacturer.
- i) Shipping Mark Guide.
- j) Installation Guide.

8.4. Each kit should be supplied with a packing note stating the quantities and descriptions of components contained in the kits, and the documents described below:

The kits should include:

Each kit should be supplied with installation instructions in Arabic and English for the offered accessories which must include the steps of installation, the number of dies, detailed drawing of



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the connector explaining the number & sequence of crimping and the width of each crimping, (Each crimping should be hexagonal).

9. SHRINKABLE TUBES DIMENSIONS

- Inner diameters of shrinkable tubes shall be large enough to pass easily through cores of cable during making joints and sizes tubes (inner /outer) should be applicable with whole joint.
- Connectors shall be according to DIN standards.
- Other dimensions shall be equal or better than the following: -

TABLE 7: Heat Shrinkable Tubes Dimensions

Item	Outer tubing (with adhesive) wall thickness / max inner diam, free recovered	Length (before shrinking)	Phase inner tubing (with adhesive) wall thickness / approx. inner diam, free recovered	Length (before shrinking)	Neutral inner tubing (with adhesive) wall thickness / inner diam, free recovered	Length (before shrinking)	Tubular braid length frees before erection
4x25 mm ²	2.5 / 55	500	2 / 22	150	2 / 22	150	400
3x35+16 mm ²	2.5 / 55	500	2 / 22	150	2 / 22	150	400
3x50+25 mm ²	2.5 / 55	750	2 / 22	150	2 / 22	150	650
3x70+35 mm ²	3.5 / 101	750	2.5 / 33	200	2 / 22	200	650
3x95+50 mm ²	3.5 / 101	750	2.5 / 33	200	2 / 22	200	650
3x120+70 mm ²	3.5 / 101	900	2.5 / 33	250	2.5 / 33	250	800
3x150+70 mm ²	3.5 / 101	900	2.5 / 40	250	2.5 / 33	250	800
3x185+95 mm ²	3.5 / 124	900	2.5 / 40	250	2.5 / 33	250	950
3x240+120 mm ²	3.5 / 124	900	2.5 / 40	250	2.5 / 33	250	1000
3x300+150 mm ²	3.5 / 124	900	2.5 / 40	300	2.5 / 40	300	1000
3x400+240 mm ²	4 / 150	1000	2.5 / 55	350	2.5 / 40	350	1000
3x500+240 mm ²	4 / 150	1000	2.5 / 80	350	2.5 / 40	350	1000



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10. TESTING AND INSPECTION

Unless otherwise accepted in writing by ... EDC, ... EDC shall require its representative(s) to witness type test. The contractor shall provide at his expense for all testing requirements. The testing dates shall be advised by the contractor to permit witnessing by ... EDC representative(s), and this must not relieve contractor of his full responsibility towards supplies guarantee. Testing of the joints should meet the requirements described in the IEC, DIN, ASTM and ISO.

Each joint should verify the following type tests: -

1-routine Tests: -

- Visual inspection.
- Dimensions.
- Quantity check.
- Kit content check.
- Connector's conductivity.

2- Type tests: -

- a) Insulation resistance (between conductor and grounded water bath 1000 M.Ohm).
- b) A.C. withstands voltage 4 KV for 15 minutes (with no break down and no flashover).
- c) Impulses withstand voltage 10+ve and 10-ve 1.2 / 50 μ s 8 KV peak between conductor and grounded water bath (with no break down and no flashover).
- d) D.C. withstand voltage 15 KV for 5 minutes (with no break down and no flashover).
- e) Electrical resistance of connectors (crimped on the cable) as per IEC 60228 table 3 and table 4 according to connectors type.
- f) Heat shrinkable material must comply with applicable standards ASTM 792, 2240, 573 and ISO 527, 868. & Ferrules shall be to DIN standard as following table (8).



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TABLE 8: Heat Shrinkable Material Testing Types

Properties	Test Method	Value (min)	Unit		
PHYSICAL					
Density	ASTM D 792 or ISO 1183,A	0.93	gm/cm3		
Tensile Strength	ISO 527-2TIP 5A or ISO 37	10	Mpa		
Ultimate Elongation	ISO 527-2TIP 5A or ISO 37	350	%		
Shore Hardness (D) 3S	ASTM D 2240 or ISO 868	40			
THERMAL AGEING					
7 DAYS @ 150 °C	ASTMD 573	TS	10	Mpa	
		EB	350	%	
PERCENTAGE CHANGE		ΔTS	-13.92	%	
		ΔEB	-11.23	%	
LOSS OF MASS		%	-0.38	%	

3- special tests: -

- Shrink of heat shrink tubes and check after free recovery (used sample from the order).
Tests and test program should be as required by EDC. Test certificates should be supplied to EDC, signed by its representative(s), shipment shall be only after acceptance of EDC representative(s).

11. The bidder shall submit with the offer :

- Copies of recently conducted type test certificates in accordance to type tests of this description.
- Catalogues and installation instructions of offered joints.
- Complete technical details with dimensional drawings including for: -



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- 1) Shrinkable tubes: inner diameters and length before shrinking and inner diameters and thickness recovered.
- 2) Connectors: length, inner and outer diameters.
- 3) Tubular braid: length, weight.

D. One complete sample kits of each size of joint.

12. GUARANTEE:

The contractor shall guarantee supplies against all defects arising out of faulty design or workmanship or defective material for a period of two year from date of delivery to ... EDC stores, inspection and technical acceptance.

The supplier shall guarantee the uniformity of the products delivered with the approved samples.



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GUARANTEE TABLE

Item	Description	Joint for cablex.....+.... mm ²
1	Name of manufacturer	
2	Country of manufacturer	
3	Type Catalogue no.	
4	Period for which this accessory has been supplied for service conditions similar to Egypt years	
5	Max design voltage kv	
6	Outer tubing : - Inner diameter /Length (before shrinking) mm/mm - Wall thickness / inner diameter free recovered mm/mm - With or without adhesive	
7	Phase inner tubing : - Inner diameter /Length (before shrinking) mm/mm - Wall thickness / inner diameter free recovered mm/mm - With or without adhesive	
8	Neutral inner tubing - Inner diameter /Length (before shrinking) mm/mm - Wall thickness / inner diameter free recovered mm/mm - With or without adhesive	
9	Tubular braid length and cross section area cm / mm ²	
10	Phase connector: length/outer dia./ inner dia. mm/mm/mm	
11	Neutral connector: length/outer dia./ inner dia. mm/mm/mm	
12	The tolerance in the inner diameter of connector	
	Phase: ± mm	
	Neutral: ± mm	

Signature -----